AELL 101
Lecture Notes

Group 12

March 4th-March 6th

Sumedh Nitin Jamsandekar
Aaron Binoj Amacadu
Abhimanyu Prakash
Naman Rindani
Sarvesh Saravanan

Department of Electrical Engineering

Indian Institute of Technology Delhi-Abu Dhabi

Lecture 1: March 4th 2025

1 AC Impedance

1.1 Resistor

Consider a resistor of resistance R (ohms, Ω) connected to an AC voltage supply V (volts, V) conducting current I (amperes, A).

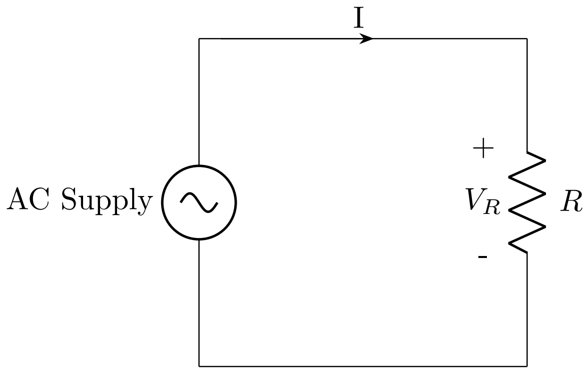


Figure 1: Circuit with a resistor showing voltage across it

The voltage across the resistor is:

$$V = V_m \cos(\omega t) = \frac{V_m \angle 0^\circ}{\sqrt{2}} \tag{1}$$

where V_m (V) is the peak voltage, $\omega = 2\pi f$ (rad/s) is the angular frequency, and t (s) is time.

The current I (A) through the resistor is:

$$I = \frac{V}{R} = \frac{V_m \angle 0^\circ}{R\sqrt{2}} \tag{2}$$

Since both voltage (1) and current (2) have the same phase angle, there is no phase difference.

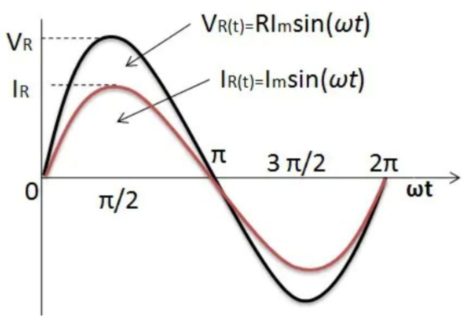


Figure 2: Sinusoidal graph of Voltage (V) and Current (A) in a purely resistive AC circuit.

Hence, the impedance Z (Ω) of a resistor in an AC circuit is:

$$Z = R \tag{3}$$

For a purely resistive circuit, voltage and current are in phase ($\theta_V - \theta_I = 0^\circ$), so the power factor is:

$$\text{pf} = \cos(0^\circ) = 1$$

This means all supplied power is real power.In all cases, pf has no unit.

1.2 Inductor

Consider an inductor of inductance L (Henries, H) connected to an AC voltage supply V (Volts, V) conducting current I (Amperes, A).

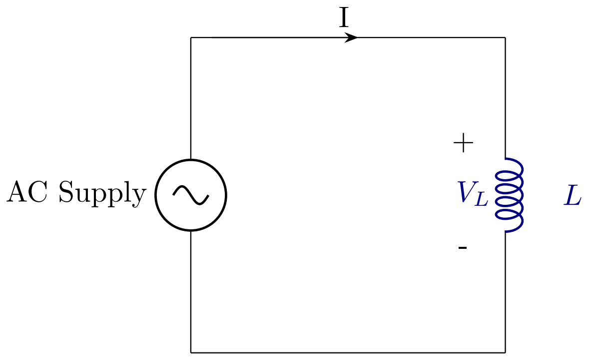


Figure 3: Circuit with an inductor showing voltage across it

The current through the inductor is:

$$I = I_m \cos(\omega t) = \frac{I_m \angle 0^\circ}{\sqrt{2}} \quad \text{(A)} \tag{4}$$

Using (4), we find the voltage across the inductor:

$$\begin{aligned} V_L &= L\frac{dI}{dt} \quad \text{(V)} \\ &= L\frac{d(I_m \cos(\omega t))}{dt} \quad \text{(V)} \\ &= -LI_m\omega \sin(\omega t) \quad \text{(V)} \end{aligned}$$

To convert this equation into phasor form, we express $\sin(\omega t)$ as its equivalent cosine form:

$$\sin(\omega t) = \cos(\omega t + 90^\circ)$$

$$\therefore V_L = LI_m\omega \cos(\omega t + 90^\circ) = \frac{L\omega I_m \angle 90^\circ}{\sqrt{2}} \quad \text{(V)} \tag{5}$$

Hence in a purely inductive AC circuit, voltage leads current by 90°.

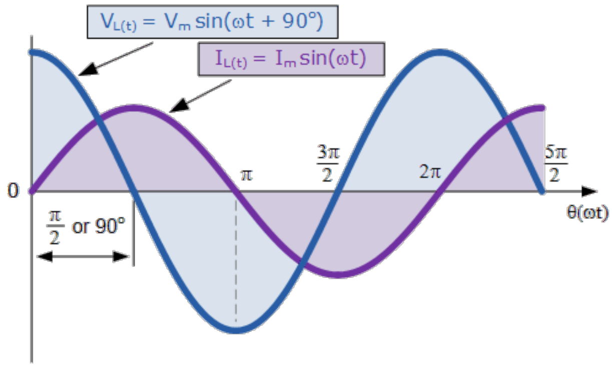


Figure 4: Sinusoidal graph of Voltage (V) and Current (A) in a purely inductive AC circuit.

The impedance of the inductor is calculated to be,

$$
\begin{aligned}
X_L &= \frac{V}{I} = L\omega\angle 90^\circ \quad (\text{Ohms, } \Omega) \\
X_L &= 2\pi f L\angle 90^\circ \quad (\Omega) \\
X_L &= j\omega L \quad (\Omega)
\end{aligned} \tag{6}
$$

where X_L is the impedance of the inductor, commonly known as **Inductive reactance**, ω is the angular frequency (rad/s), f is the AC frequency supply in hertz (Hz), and $j = \sqrt{-1}$.

For a purely inductive circuit, current lags voltage by 90°, so the power factor is:

$$\text{pf} = \cos(90^\circ) = 0$$

This means no real power is consumed; all power is reactive.

1.3 Capacitor

Consider a capacitor of capacitance C (in farads, F) connected to an AC voltage supply V (in volts, V) conducting current I (in amperes, A).

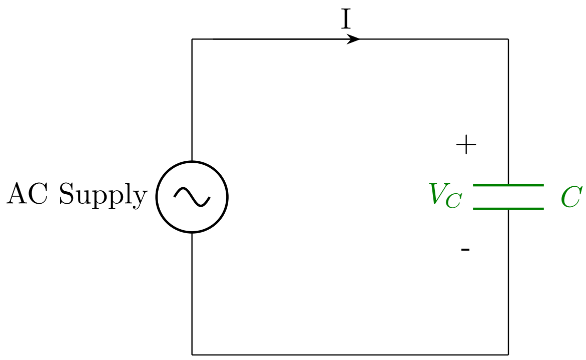


Figure 5: Circuit with a capacitor showing voltage across it

The current through the capacitor is:

$$I = I_m \cos(\omega t - 90^\circ) = \frac{I_m\angle -90^\circ}{\sqrt{2}} \quad (\text{A}) \tag{7}$$

Using (7), we can calculate the voltage across the capacitor:

$$
\begin{aligned}
V_C &= \frac{1}{C}\int I\,dt \quad (\text{V}) \\
&= \frac{1}{C}\int I_m \cos(\omega t - 90^\circ)\,dt \\
&= \frac{I_m}{\omega C}\sin(\omega t - 90^\circ) \quad (\text{V})
\end{aligned}
$$

To express this equation in phasor form, we use the identity:

$$\sin(\omega t - 90^\circ) = \cos(\omega t)$$

$$\therefore V_C = \frac{I_m}{\omega C}\cos(\omega t) = \frac{I_m \omega C\angle 90^\circ}{\sqrt{2}} \quad (\text{V}) \tag{8}$$

Hence, in a purely capacitive AC circuit, the voltage lags the current by 90°.

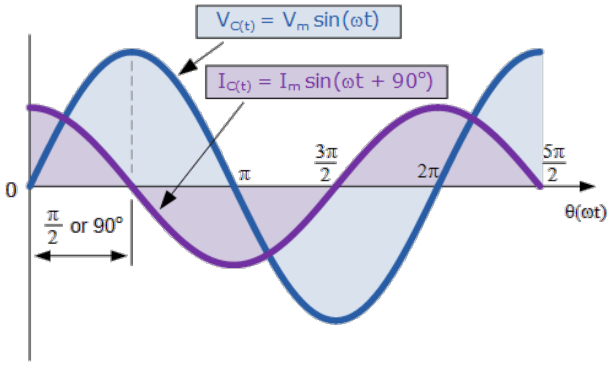


Figure 6: Sinusoidal graph of Voltage (V) and Current (A) in a purely capacitive AC circuit.

The impedance of the capacitor is calculated as:

$$\begin{aligned} X_C &= \frac{V}{I} = \frac{1}{j\omega C} \quad (\text{ohms}, \Omega) \\ X_C &= \frac{1}{2\pi f C} \angle -90° \quad (\Omega) \\ X_C &= \frac{1}{j\omega C} \quad (\Omega) \end{aligned} \tag{9}$$

where X_C is the impedance of the capacitor, commonly known as **Capacitive reactance**, ω (rad/s) is the angular frequency, f (Hz) is the AC frequency supply in hertz, and $j = \sqrt{-1}$.

For a purely capacitive circuit, current leads voltage by 90°, so the power factor is:

$$\text{pf} = \cos(-90°) = 0$$

This means no real power is consumed; all power is reactive.

2 Relation of Reactance and frequency for Inductors and Capacitors

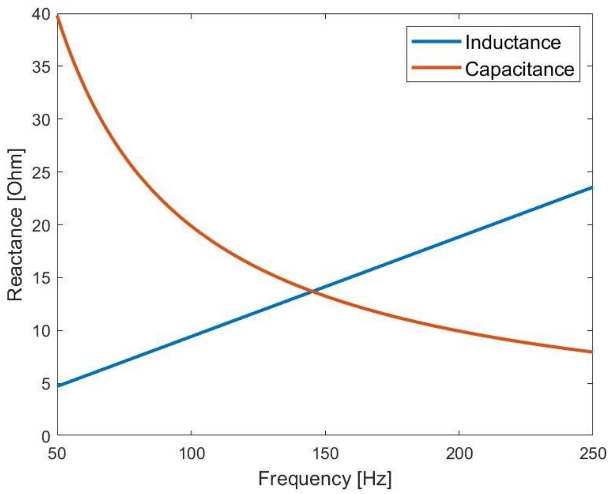
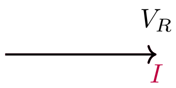


Figure 7: Graph of Reactance Vs Frequency for Inductance and Capacitance.

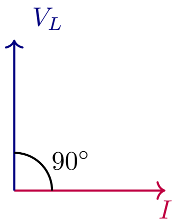
From 9 and 6 we can conclude that Inductive reactance is linearly proportional to Frequency while Capacitive reactance is inversely proportional to Frequency.

3 Phasor Diagram for the three devices

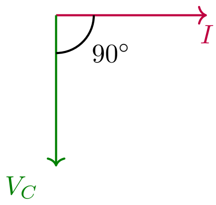
Using 1,5,8, we can plot the phasor diagrams for the resistor, inductor, and capacitor, respectively for current, I at 0°.



Resistance (In-phase)



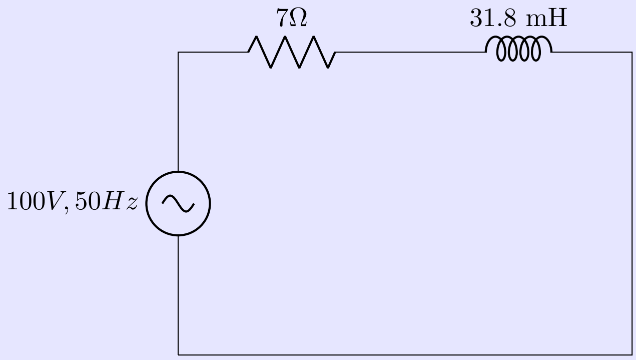
Inductance (+ 90°)



Capacitance (- 90°)

Problem Statement

A resistance of 7Ω is connected in series with an inductor of pure inductance of 31.8 mH and is connected to a $100V$, $50Hz$ AC supply. Calculate circuit current and phase angle between V and I.



Solution

Step 1: Calculate Inductive Reactance

Using (eq 6),

$$Z_L = 2\pi f j L = 2\pi(50)(31.8 \times 10^{-3}) = 10j\Omega$$

Thus, the impedance is given by:

$$Z = (7 + j10)\,\Omega$$

Step 2: Calculate Circuit Current in Rectangular Form

$$I = \frac{100}{7 + j10}$$

Multiply by the conjugate:

$$I = \frac{100(7 - j10)}{(7 + j10)(7 - j10)} = \frac{700 - j1000}{149}$$

$$I = 4.7 - j6.7 \quad \text{(A)}$$

Step 3: Convert Current to Polar Form

$$|I| = \sqrt{(4.7)^2 + (-6.7)^2} = \sqrt{66.98} = 8.2A$$

$$\theta = \tan^{-1}\left(\frac{-6.7}{4.7}\right) = \tan^{-1}(-1.4255) = -54.95^\circ$$

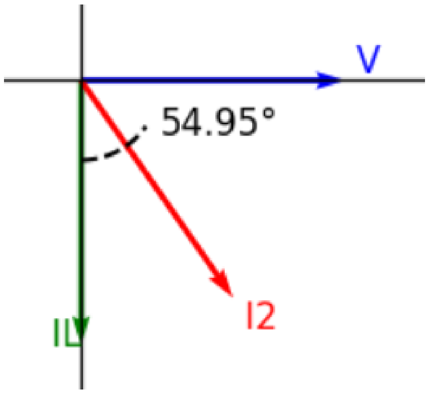


Figure 8: Impedance

Since the imaginary part is negative, the current lags voltage.

Final Answer

- **Rectangular Form:** $I = 4.7 - j6.7$ A
- **Polar Form:** $I = 8.2\angle - 54.95^\circ$ A
- **Phase Angle:** $\theta = -54.95^\circ$ (Current lags voltage)

Lecture 2: March 5th, 2025

1 AC Power

The voltage and current in AC circuits are given by:

$$V(t) = V_m \cos(\omega t + \theta_V) \quad \text{(V)} \tag{10}$$
$$I(t) = I_m \cos(\omega t + \theta_I) \quad \text{(A)} \tag{11}$$

where:

- V_m is the peak voltage in volts (V),
- I_m is the peak current in amperes (A),
- ω is the angular frequency in radians per second (rad/s),
- θ_V and θ_I are the phase angles of voltage and current in radians.

The instantaneous power is:

$$P(t) = V(t)I(t) \quad \text{(W)}$$

Expanding this,

$$P(t) = \frac{1}{2} I_m V_m \left[\cos(\theta_V - \theta_I) + \cos(2\omega t + \theta_V + \theta_I)\right] \quad \text{(W)}$$

Since the term $\cos(2\omega t + \theta_V + \theta_I)$ averages to zero over a complete cycle, the average power is:

$$P_{\text{avg}} = \frac{1}{2} I_m V_m \cos(\theta_V - \theta_I) \quad \text{(W)} \tag{12}$$

For an inductor, $\theta_V - \theta_I = \frac{\pi}{2}$, hence the average power is zero watts (W).

Problem Statement

For given values $L = 0.1\,H$, $C = 0.32\,mF$, and $R = 10\,\Omega$, find the net impedance and the current, I.

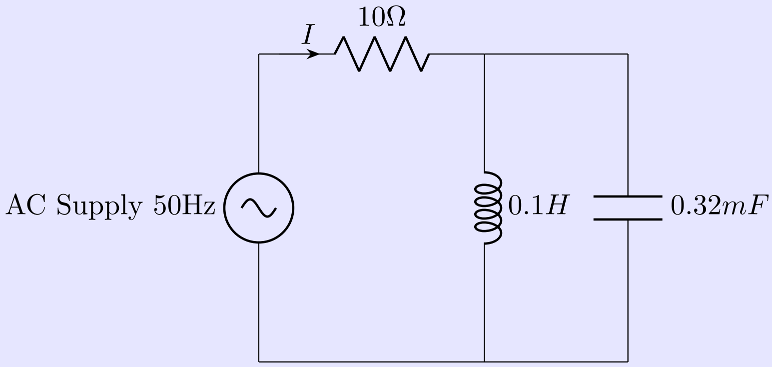


Figure 9: RLC Circuit Diagram

Step 1: Calculate Inductive Reactance and Capactive Reactance

$$Z_L = j(2\pi \times 50) \times 0.1\ ,\Omega = 31.4j\,\Omega$$
$$Z_C = \frac{1}{j\omega C} = -\frac{j}{2\pi \times 50 \times C} \approx -j(10)\,\Omega$$

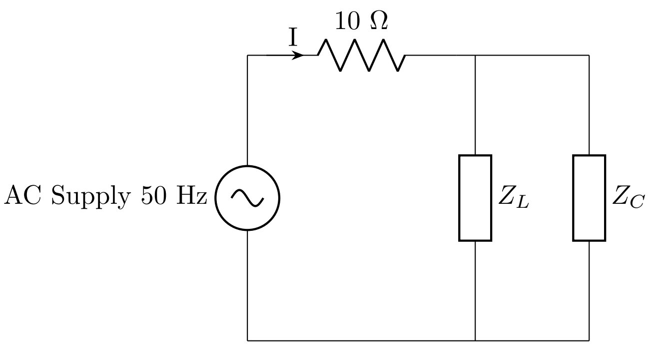


Figure 10: Equivalent Circuit

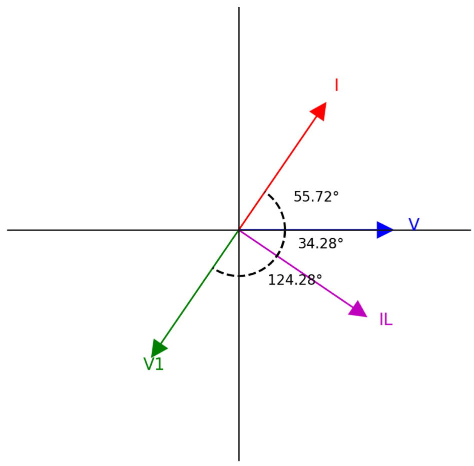


Figure 11: Current Phase Analysis

The net impedance is:

$$\begin{aligned}
Z_{eq} &= 10 + \frac{1}{\left(\frac{1}{31.4j} - \frac{1}{10j}\right)}\,\Omega \\
&= 10 - 14.67j\,\Omega \\
&= \sqrt{10^2 + 14.67^2}\angle \tan^{-1}\left(\frac{-14.67}{10}\right)\,\Omega \\
&= 17.75\angle -55.72^\circ\,\Omega
\end{aligned}$$

Step 2: Calculate Circuit Current in Rectangular Form and Convert to Polar Form

$$
\begin{aligned}
I = \frac{V}{Z_{eq}} &= \frac{100\angle 0^\circ}{17.75\angle -55.72^\circ}\, A \\
&= 3.173 + 4.655j\, A \\
&= 5.63\angle \tan^{-1}\left(\frac{4.655}{3.173}\right)\, A \\
&= 5.63\angle 55.72^\circ\, A
\end{aligned}
$$

Final Answers

- **Net Impedance:** $Z_{eq} = 10 - 14.67j\,\Omega$
- **Current (Rectangular Form):** $I = 3.173 + 4.655j\, A$
- **Current (Polar Form):** $I = 5.63\angle 55.72^\circ\, A$
- **Phase Relationship:** Current leads voltage by 55.72°.

Lecture 3: March 6th, 2025

1 AC Power

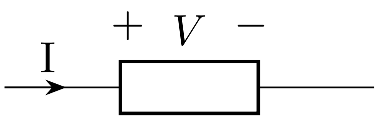


Figure 12: Voltage (V) and current (A) across a device

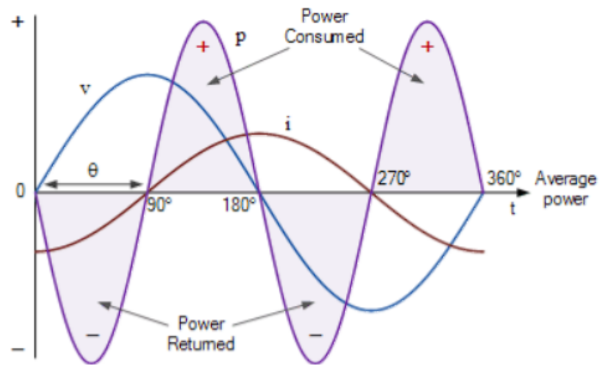


Figure 13: AC Power graph (W)

- Instantaneous power formula: $p(t) = v(t) \cdot I(t)$ (W).
- Average power calculation: $P_{\text{avg}} = \frac{1}{2} V_m I_m \cos(\theta_V - \theta_I)$ (W).
- Power factor as $\cos(\theta)$, where θ is the phase angle difference between voltage and current.

2 Voltage and Current Representation

$$V = \frac{V_m}{\sqrt{2}}\angle\theta_V = V_{\text{rms}}\angle\theta_V \quad \text{(V)} \tag{13}$$

$$I = \frac{I_m}{\sqrt{2}}\angle\theta_I = I_{\text{rms}}\angle\theta_I \quad \text{(A)} \tag{14}$$

3 Average Power Calculation

$$P_{\text{avg}} = V_{\text{rms}} I_{\text{rms}} \cos(\theta_V - \theta_I) \quad \text{(W)}$$

where $\cos(\theta_V - \theta_I)$ is known as the **power factor**.

We define the apparent power as:

$$S = \mathbf{V} \cdot \mathbf{I}^*$$

where $\mathbf{I}^*$ is the conjugate of $\mathbf{I}$. Expanding this,

$$S = V_{\text{rms}} I_{\text{rms}} \angle(\theta_V - \theta_I) \quad \text{(VA)}$$

$$S = V_{\text{rms}} I_{\text{rms}} [\cos(\theta_V - \theta_I) + j \sin(\theta_V - \theta_I)] \quad \text{(VA)} \tag{15}$$

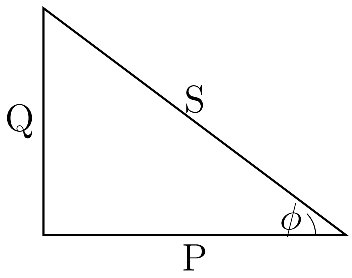
4 Real and Reactive Power

$P = V_{\text{rms}} I_{\text{rms}} \cos(\theta_V - \theta_I)$ (Real Power (W) – Actual power consumed by the circuit)

$Q = V_{\text{rms}} I_{\text{rms}} \sin(\theta_V - \theta_I)$ (Reactive Power (VAR) – Power oscillating between source and reactive elements)

$S = P + jQ$ (Apparent Power (VA) – Total complex power in the circuit)

5 Power Triangle



A useful representation of power components is the power triangle:

$$P = S \cos\phi \quad \text{(W)}$$
$$Q = S \sin\phi \quad \text{(VAR)}$$
$$S = \sqrt{P^2 + Q^2} \quad \text{(VA)}$$

6 Power Factor and Impedance

The power factor (pf) is given by:

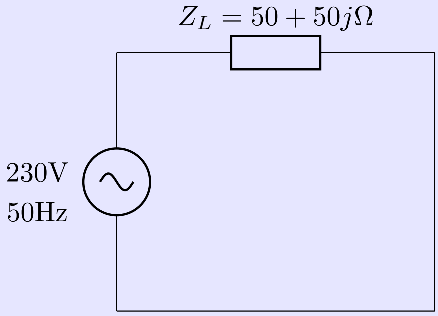
$$\text{pf} = \frac{\text{Real Power (W)}}{\text{Apparent Power (VA)}} = \frac{P}{S} \tag{16}$$

For an impedance $Z = R + jX$, we have:

$$S = I_{\text{rms}}^2 Z \quad \text{(VA)} \tag{17}$$
$$P = I_{\text{rms}}^2 R \quad \text{(W)} \tag{18}$$
$$Q = I_{\text{rms}}^2 X \quad \text{(VAR)} \tag{19}$$

Problem Statement

For a given circuit of AC voltage of 230V,50Hz, supplied to an impedence of 50+50j, find the real and reactive power of the element.



$$i = \frac{V}{Z_{eq}} = \frac{100V}{(50+50j)\Omega} = \frac{2}{1+j} = (1-j)\text{ A}$$

Step 1: Express the current in polar form

$$i = \sqrt{2}\angle\tan^{-1}(-1) = 1.41\angle-45^\circ\text{ A}$$

Step 2: Find the apparent power in the circuit (using eq: 15)

$$S = V_{rms}I_{rms}(\cos\phi + j\sin\phi)$$

where $\phi = 45^\circ$. Substituting values:

$$S = 100\sqrt{2}(1 + j\sqrt{2})\text{ VA}$$

$$S = (100 + 100j)\text{ VA}$$

Step 3: Find Real and Reactive Power (using eq 4)

$$S = P + jQ$$

we obtain:

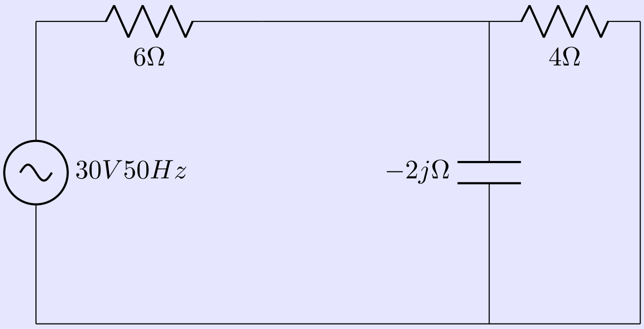
$$P = 100\text{ W} \qquad \text{(Real Power – Active Power)}$$
$$Q = 100\text{ VAR} \qquad \text{(Reactive Power)}$$

Final Answers

- **Current (Rectangular Form):** $i = (1-j)$ A
- **Current (Polar Form):** $i = 1.41\angle-45^\circ$ A
- **Apparent Power:** $S = (100 + 100j)$ VA
- **Real Power (Active Power):** $P = 100$ W
- **Reactive Power:** $Q = 100$ VAR

Problem Statement

For the given circuit, find the power factor and the average power delivered across the circuit.



Step 1: Find Equivalent Impedance of the Circuit

$$\begin{aligned} Z_{eq} &= (-2j\,\Omega || 4\,\Omega) + 6\,\Omega \\ &= \left(\frac{-1}{2j} + \frac{1}{4}\right)^{-1} + 6 \\ &= \left(\frac{4}{2j+1}\right) + 6 \\ &= \left(\frac{10+12j}{2j+1}\right)\Omega \\ &= 7\angle -13.24^\circ\,\Omega \end{aligned}$$

Step 2: Calculate the Current Across the Circuit

$$\begin{aligned} i &= \frac{V}{Z_{eq}} = \frac{30}{7}\angle(0^\circ - (-13.24^\circ)) \\ &= 4.28\angle 13.24^\circ \text{ A} \end{aligned}$$

Step 3: Calculate the Power Factor and the Power Delivered

Power factor:

$$\text{pf} = \cos(\theta_V - \theta_I) = \cos(-13.24^\circ) = 0.973 \quad \text{(leading)}$$

Power delivered:

$$\begin{aligned} P &= V_{rms} I_{rms} \cos\phi \\ &= 30 \times 4.28 \times 0.973 \text{ W} \\ &= 124.93 \text{ W} \approx 125 \text{ W} \end{aligned}$$

Final Answers

- **Equivalent Impedance:** $Z_{eq} = 7\angle -13.24^\circ\,\Omega$
- **Current (Polar Form):** $i = 4.28\angle 13.24^\circ$ A
- **Power Factor:** 0.973 (Leading)
- **Power Delivered:** $P = 125$ W

Note: If $\phi > 0$, the power factor is lagging; if $\phi < 0$, it is leading. A high power factor (0.97 to 0.99) is considered efficient.

7 Group Contributions

- **Sumedh Nitin Jamsandekar:** Worked on Lecture 1 notes and overall LaTeX editing.
- **Aaron Binoj Amacadu:** Worked on graphs and circuit diagrams.
- **Abhimanyu Prakash:** Worked on Lecture 2 theory notes.
- **Naman Rindani:** Worked on Lecture 3 theory notes.
- **Sarvesh Saravanan:** Worked on Lecture 3 problem statements.